

accumulation of P factor in a muscle.

- Angina is relieved by rest because this decreases the myocardial O₂ requirement and permits the blood supply to remove the factor. Intermittent claudication, the pain produced in the leg muscles of persons with occlusive vascular disease, is another example.
- It characteristically comes on while the patient is walking and disappears on stopping.
- Visceral pain, like deep somatic pain, initiates reflex contraction of nearby skeletal muscle.
- This reflex spasm is usually in the abdominal wall and makes the abdominal wall rigid. It is most marked when visceral inflammatory processes involve the peritoneum. However, it can occur without such involvement.
- The spasm protects the underlying inflamed structures from inadvertent trauma. Indeed, this reflex spasm is sometimes called “guarding.”

4. Stretch reflex (I)

STRETCH (Myotatic) REFLEX (figure 20)

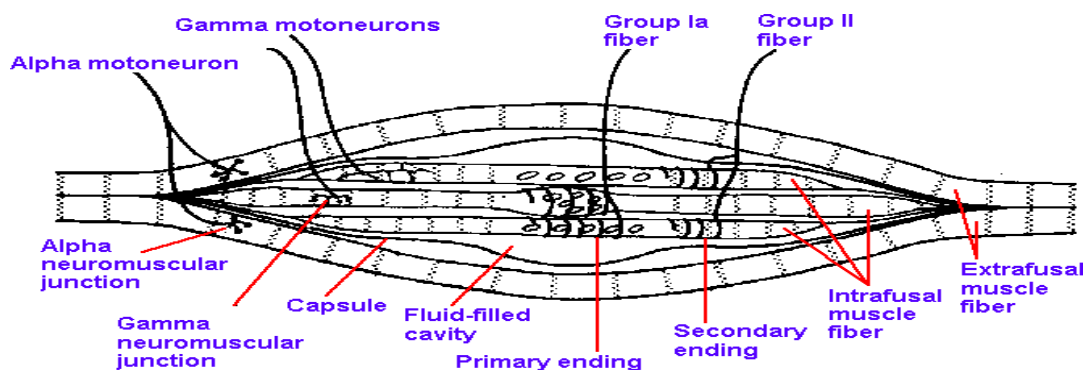
- Stretch of any skeletal muscle produces reflex contraction of the stretched muscle to oppose the stretch (increased length).
- It is a reflex contraction of a muscle when it is passively stretched.
- Receptors: the muscle spindle
 - Pathway

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1. Stimulus: → passive stretch
2. Receptors: → ms spindles
3. Afferents: → fast-conducting A α (Ia) and A β (II) nerve fibers
4. Centers: → alpha motor neurons in AHCs.
5. Efferents: → axons of alpha motor neurons.
6. Effector organ: → Extrafusal ms

fibers The muscle spindle (figure 20)

- It is a stretch receptor belonging to proprioceptors.
- It is formed of a connective tissue sheath enclosing 2 muscle fibers (intrafusal fibers); parallel to the extrafusal contractile muscle fibers of the skeletal muscle and both are attached to the muscle tendon at the origin and insertion.
- The intrafusal muscle fibers are fusiform and less developed than the extrafusal muscle fibers.



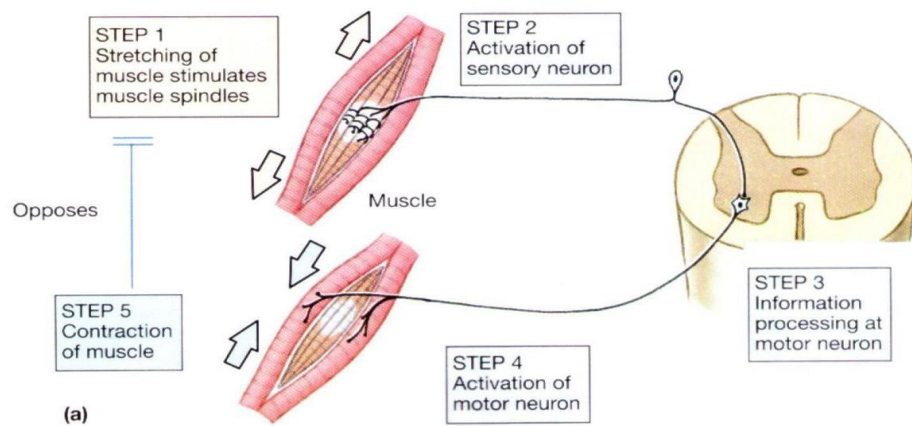


Figure 20: the muscle spindle and Stretch reflex)

- Each intrafusal fiber has a central non-contraction stretchable part containing nuclei and a peripheral contractile part containing less developed actin and myosin.
- According to the shape of the central receptor area, two types of intrafusal muscle fibers are present (figure 21):
 1. Nuclear bag fibers; the central part is a bag full of nuclei.
 2. Nuclear chain fibers; the central part contains few nuclei arranged as chain.
- 3. Nerve supply of muscle spindle:
 - Afferent sensory fibers: They end by two types of afferent endings on the central receptor area of the Spindle:
 - Primary endings: annulospiral, - coiling around nuclear bag and nuclear chain fibers. Afferent fibers are - thick myelinated, 16 in diameter (Ia) fibers - Conducting at 70-120 m/second. Rapidly adapting receptors (respond to the rate of change in length).

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Secondary endings:

- flower spray, - only in nuclear chain on either side of primary ending.
- Afferent fibers are – thin myelinated, - 8 μ in diameter (Ib,II) fibers.
Conducting at 30- 70 m/second.
- Slowly adapting receptors (respond to new increased length).

Efferent motor fibers:

- They are called gamma (γ) efferent motor fibers.
- They supply the peripheral contractile part of the intrafusal muscle fibers.
- They are the axons of the small anterior horn cells (γ AHCS) present in the anterior horn grey matter together with the large anterior horn cells (α AHCS) that supply the extrafusal muscle fibers.
- They are thin myelinated (4 μ in diameter), while the α motor fibers are thick myelinated. Both run together in the motor nerve supply of any skeletal muscle (30% γ efferents and 70 % α efferents).
- γ efferents end on:
 - the nuclear bag fibers by plate endings (motor end plates) $\rightarrow$ dynamic γ efferents.
 - the nuclear chain fibers by trail endings (extensive network) $\rightarrow$ static γ efferents.

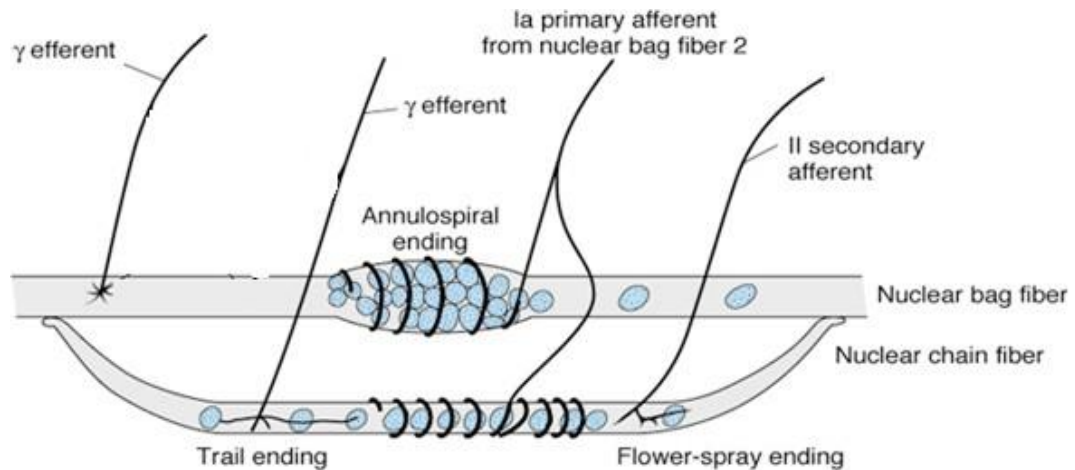


Figure 21: intrafusal fibers

Mechanism of spindle stimulation

The stimulus is stretch (increased length) of the muscle spindle. This may be produced as follows:

1) Stretch of the whole muscle by:

- Anatomical; the distance between the origin and insertion of any muscle is longer than the exact length of the muscle fiber.
- Contraction of a group of muscles (agonists) will stretch the antagonist muscles.
- Tapping on the muscle tendon with a hammer.
- Pulling on the muscle tendon in the direction of insertion.

Stretching the whole muscle stretches the central receptor area of the spindle → stimulates the spindle → ↑ afferent discharge.

2) γ efferent discharge even without change in muscle length → contraction of the peripheral parts of intrafusal muscle fibers → stretches the central receptor area of the spindle → stimulates the spindle.

3) Maximal stimulation and discharge occurs when the muscle is stretched (1) + ↑ γ efferent discharge (2).

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4) Muscle contraction without $\uparrow \gamma$ efferent discharge $\rightarrow$ muscle shortening $\rightarrow$ the central receptor area of the spindle also shortens $\rightarrow$ decreasing spindle discharge.

5) α - γ link: (simultaneous discharge of α and γ efferents) muscle contraction (due to α AHC discharge) with increased γ efferent discharge $\rightarrow$ muscle shortening including the intrafusal muscle fibers $\rightarrow$ however, the central receptor area of the spindle is stretched by contraction of the peripheral part due to increased γ efferent discharge $\uparrow$ sensitivity of the spindle to stretch even during contraction and shortening of the muscle.

6) The primary (dynamic annulospiral) endings are stimulated during the act of stretch of the muscle and stop to discharge when the stretch is stopped even if the length of the muscle is increased $\rightarrow$ respond by tendon jerk.

The stimulus is the rate of stretch ($\Delta L/\Delta t$, where ΔL = change in length and Δt = change in time).

7) The secondary (static, flower spray) endings are stimulated by the stretched (increased length) muscle respond by muscle tone.

The stimulus is the new increased length (ΔL).

Types of Stretch Reflex (table 8)

	Dynamic Response	Static Response
Stimulus	Sudden stretch	Maintained (steady) stretch
Receptors	nuclear bag	nuclear chain
Afferents	1ry endings	primary and secondary endings
Center	Alpha motor neurons	Alpha motor neurons

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Response	Rapid contraction followed by rapid relaxation	Maintained subtetanic contraction
Examples	e.g. tendon jerk	e.g. muscle tone

Functions of the stretch reflex

(1) Muscle tone:

- It is the static response of the stretch reflex. Mechanism:
 - All muscles in the body are in a state of maintained stretch because the distance between the origin and insertion is longer than the exact length of the muscle fiber → continuous stimulation of the secondary receptors maintained contraction (muscle tone).
- Functions:
1. Muscle tone is higher in the antigravity muscles to keep the erect posture of the body against the force of gravity. Variations of muscle tone in the antigravity muscles is mediated by the postural reflexes.
 2. It helps venous and lymph return from the lower limbs against the force of gravity.
 3. It is responsible for basal heat production (basal metabolic rate) to maintain body temperature.

Tone of the abdominal muscles keeps the viscera in position and prevents Visceroptosis.

(2) The load reflex:

- Stretch of the muscle by a load → reflex feed back contraction of the muscle to oppose the stretch, support the load and maintain the limb in a fixed position.

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- Significance:

It is important to keep a limb in a fixed position and resist any external force tending to move it.

Example:

* the position of the limb during threading a needle. * the forearm keeps the horizontal position although a load is put in the hand.

Mechanism:

Any voluntary motor function involves:

discharge of α AHCs to extrafusal muscle fibers $\rightarrow$ muscle contraction to put The limb in the desired position.

- discharge of γ AHCs to intrafusal muscle fibers $\rightarrow$ increasing the sensitivity of the spindle.

Putting a load in the arm $\rightarrow$ stretches the supporting muscles by its weight $\rightarrow$ stretches the spindle $\rightarrow$ increases the afferent discharge - $\rightarrow$ more stimulation of α AHCs $\rightarrow$ stronger contraction of the supporting muscle to prevent falling of the load and keeping the fixed position of the limb.

(3) The damping function:

- The α AHC is the site of convergence of many afferents ; some are excitatory including afferents from muscle spindles and others are inhibitory. If the afferents from muscle spindles are cut $\rightarrow$ the membrane potential of α AHC oscillates continuously between:

- central excitatory state $\rightarrow$ $\uparrow$ frequency of discharge $\rightarrow$ stronger contraction, and
- central inhibitory state $\rightarrow$ $\downarrow$ frequency of discharge $\rightarrow$

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weaker contraction.

Both effects result in unsmooth contraction called Tremors

- if the afferents from muscle spindles are intact → feedback stabilization of the α AHC membrane potential → stable frequency of motor discharge a central inhibitory state → smooth contraction. This is called damping function of the spindle.
- Mechanism: $\uparrow$ central excitatory state of α AHC → $\uparrow$ frequency of discharge stronger contraction and shortening → $\downarrow$ frequency of discharge of muscle spindle → $\downarrow$ central excitatory state, and vice versa.

(4) The servo-assistant function:

- In any voluntary motor function, the cerebral cortex sends descending motor impulses to
 1. α AHCs → motor discharge to extrafusal muscle fibers → muscle contraction → shortening → decreasing the sensitivity of the spindle and,
 2. γ AHCS to intrafusal muscle fibers increasing the sensitivity of the spindle.
- The sensitivity of the muscle spindle during contraction depends on:
 1. Muscle contraction → shortening of the spindle → $\downarrow$ sensitivity.
 2. The force applied to the muscle → determines the type of muscle contraction (isometric or isotonic) → and consequently the degree of shortening.
 3. γ AHC discharge → $\uparrow$ the sensitivity of the spindle.

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- During isometric contraction (e.g. to lift a weight that is too heavy to be carried) → no shortening occurs → increased sensitivity of the spindle by γ AHC discharge not antagonized by shortening → increased afferent discharge from the spindle to α AHCs → additional reflex motor discharge → stronger contraction to lift the heavy weight independent of the cerebral cortex. This is the assistant function of the stretch reflex to the motor function of the cerebral cortex.

Supraspinal Centers Controlling The Stretch Reflex

- Both the α AHCS and γ AHCS receive continuous facilitatory and inhibitory impulses from the supraspinal centers along the descending tracts.
- The net effect is excitatory under normal condition.
- This supraspinal control allows variations in muscle tone to maintain posture.

The supraspinal facilitatory centers:

1. The vestibular nuclei: Site: in the medulla oblongata. Pathway: - along the lateral and ventral vestibulospinal tracts. - The facilitatory effect is ipsilateral. Significance: Its action is more marked in decerebrate rigidity.
2. The facilitatory reticular formation: Site: In the whole brain stem. Structure: Network of intercommunicating neurones forming reticular nuclei and fibers. The neurons are of two types: sensory small neurons with short axons and multiple connections forming neuronal pool. They receive impulses from all ascending tracts. * Large motor neurones that receive impulses from the sensory neurones. form ventral reticulospinal tracts that end on Their axons either: A- descend spinal cord motor neurons. Or

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B- ascend from the reticular activating system 09 The Central Nervous System con-specific thalamic nuclei and all

areas of cerebral cortex → uo puə e maintains wakefulness and consciousness. significance: sleep and in abnormal cases → coma. Loss of activity It has intrinsic activity independent of other facilitatory centers.

3. The cortical facilitation: Cita: Motor area 4 and other areas in the parietal lobe. Pathway: - along the pyramidal tract (contralateral). Significance: Section of the pyramidal tract in the pyramid flaccid paralysis. inhibition of the stretch reflex
4. The neocerebellum: Site: Neocerebellum. Pathway: . No direct pathway between the neocerebellum and the spinal cord motor - The neocerebellum exerts its facilitatory effect on the spinal cord motor centers indirectly by facilitating the other supraspinal facilitatory centers. - The facilitatory effect is ipsilateral. centers.

The supraspinal inhibitory centers:

1. The cortical inhibition: Site: Premotor area 6 Pathway: - along the extrapyramidal tracts mainly the lateral reticulospinal. Significance: Extrapyramidal lesions → exaggerated stretch reflex.
2. The basal ganglia
3. The paleocerebellum
4. The inhibitory reticular formation: Site: In the ventromedial medulla oblongata (bulboreticular formation). It is stimulated by descending impulses from the other inhibitory centers and relays them to the spinal cord motor centers by the lateral ticulospinal tract. Pathway: - along the

the lateral reticulospinal tract. Significance: It has no intrinsic activity.

5. Stretch reflex (II)

Clinical Applications of the Stretch Reflex

(1) The dynamic response:

A- The tendon jerk (figure 22, 23):

- Definition: Sudden stretch of the muscle by tapping on its tendon → reflex rapid contraction of the muscle followed by rapid relaxation.
- Mechanism: Sudden stretch of the muscle → stimulation of primary endings of the muscle spindle → ↑ afferent discharge from the spindle → enter the spinal cord along the dorsal roots and end directly on the αAHCs → ↑ efferent discharge to extrafusal muscle fibers → reflex contraction and shortening → ↓ or stop afferent discharge from the spindle → muscle relaxation.

Examples (table 9):

Reflex	Description	Center
The biceps reflex	Tapping on the biceps tendon → reflex contraction of the biceps and flexion of the elbow.	C4- C5
The triceps reflex	Tapping on the triceps tendon → reflex contraction of the triceps the and extension of elbow	C5 - C6
The knee jerk	Tapping on the patellar tendon → contraction of the quadriceps and extension of the knee	L3 - L4
The ankle jerk	Tapping on the Tendon Achillis → reflex contraction of the muscle gastrocnemius and plantar flexion of the ankle	S1- S2